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Modeling and Prediction of Water Quality Using Artificial Intelligence Techniques for Enhanced Monitoring and Management

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ABSTRACT

Artificial intelligence offers a cost-effective solution for water supply and sanitation systems while ensuring compliance with quality standards. This study presents a novel AI-based system for real-time monitoring and prediction of water quality, with a focus on sustainability and environmental protection. Using adaptive regression models, the water quality index (WQI) was predicted, and various classifiers were employed to categorize water quality classification (WQC). The dataset included 10 parameters: pH, EC, Ca^{2+}, Mg^{2+}, Na^{+}, K^{+}, CO_3^{2-}, HCO_3^{-}, Cl^{-}, and SO_4^{2-}, which are essential for building efficient prediction models. Among the models tested, linear regression (LR) performed best for predicting WQI, achieving a mean absolute error (MAE) of 2.18 and a coefficient of determination (R^2) of 0.967, whereas the K-nearest neighbors (KNN) and random forest (RF) classifiers achieved classification accuracies of 92.3% and 91.6%, respectively, for WQC. The LR model provided accurate WQI predictions, and KNN excelled in classifying WQC. An ensemble model, that combines regression and classification techniques, achieved a high regression coefficient of 96.7% for predicting both the WQI and WQC. This AI-driven approach enhances water quality prediction and supports better water treatment and management systems. These findings have practical implications for improving water quality monitoring and decision-making in pollution-affected regions such as Tamil Nadu, India.

1 | Introduction

Rapid industrialization and urban expansion in Tamil Nadu, India, have significantly deteriorated water pollution. The indiscriminate discharge of industrial effluents and unplanned urban wastewater has severely impacted surface and groundwater resources, leading to pressing public health and environmental challenges (Khanam et al. 2022). An in-depth understanding of water quality issues and patterns is crucial for effective pollution reduction and regulatory measures (Bansal and Parthasarathy 2020). Globally, nations are initiating and refining environmental water management strategies to gain comprehensive insights into aquatic ecosystems (Mishra et al. 2021). Water, a fundamental resource for life, is predominantly saline, with 95% of the Earth's water being saltwater despite covering 71% of the planet's surface (Mishra 2023). Therefore, safeguarding the quality of freshwater resources is of paramount importance. Approximately one billion individuals lack access to reliable drinking water sources, and annually, two million deaths occur because of waterborne diseases and inadequate sanitation and hygiene (Organization, W. H 2019). In Tamil Nadu, outbreaks of diseases like cholera, dysentery, and

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Summary

- AI models effectively predict irrigation water quality using lab-analyzed groundwater parameters.
- Linear regression model showed highest accuracy for water quality index prediction ($R^2 = 99\%$).
- K-nearest neighbor model best classified water quality classes ($R^2 = 90\%$).
- Ensemble modeling improved overall accuracy and reliability in water quality classification.
- Study supports AI-driven decision tools for groundwater quality monitoring in industrially impacted agricultural regions.

typhoid are often linked to contaminated water sources, emphasizing the need for robust water quality monitoring and predictive systems (Vasudevan et al. 2021).

Accurate prediction of water quality is essential for effective water management and pollution control, especially in industrial zones where contamination levels can vary significantly. Forecasting water quality helps anticipate changes and implement timely interventions to prevent severe pollution (Gazzaz et al. 2012). Traditional methods of water quality assessment, which involve labor-intensive laboratory analyses and sample testing, are often slow and costly. As industrial activities contribute to fluctuating levels of pollutants, there is a pressing need for advanced techniques to ensure timely and accurate water quality monitoring (Hu et al. 2019). Enhanced predictive models can improve the efficiency of water diversion schemes and infrastructure management, reducing the economic burden associated with water treatment and infrastructure upgrades (Liu et al. 2020). Traditional water quality assessment techniques—such as laboratory-based sampling and analysis—are resource-intensive, time-consuming, and spatially limited (Moretti et al. 2024). These methods often fail to capture the rapid and nonlinear fluctuations in pollution levels, especially in industrial zones where contaminant concentrations can vary drastically over short time scales (Jayaraman et al. 2024). Moreover, the high cost and delayed results of these traditional approaches hinder timely decision-making, posing a risk to both public health and environmental sustainability.

Advancements in artificial intelligence (AI) offer promising solutions for addressing water quality challenges in Tamil Nadu's industrial areas. AI techniques such as regression and classification models are followed by the stacking of these models to generate an ensembled model that has demonstrated considerable performance in predicting the water quality index (WQI) (Shams et al. 2024; Hmoud Al-Adhaileh and Waselallah Alsaade 2021). These models can analyze complex and nonlinear relationships within water quality data, providing more accurate forecasts than traditional methods do. By employing AI models such as regression, classification, and ensembled models it is possible to develop a real-time system for monitoring and predicting water quality (El-Kenawy 2025). This approach can enhance decision-making and strategy formulation for water pollution management, ensuring cleaner and safer water resources in industrial regions of Tamil Nadu (Tung and Yaseen 2020). The present study aims to develop a robust AI-based framework for water quality prediction and classification tailored to the specific needs of Tamil Nadu's industrial regions. The proposed model integrates multiple machine learning techniques—including regression and classification models—stacked into an ensemble architecture to accurately predict the WQI and determine water quality classes (WQC). Additionally, classification models are utilized for water quality classification (WQC), which is followed by stacking of various regression and classification models to predict WQI and WQC. These AI techniques offer a generalized framework for forecasting water pollution, thus enabling more effective and timely decision-making by stakeholders.

2 | Materials and Methods

2.1 | Data Collection and Assessment

The study area covers approximately 1200 km^2 in the Erode and Namakkal districts of Tamil Nadu, India, within the Survey of India toposheet numbers 58D/7 and 58D/8 at a 1:50,000 scale. It is located between longitudes 77°36′ E and 77°52′ E and latitudes 11°10′ N and 11°30′ N (Rao et al. 2005). The region features a diverse topography, including low hill ranges and extensive flat plains. The elevations range from over 500 m in elevated terrains to less than 500 m in predominantly flat areas, which are characterized by extensive agricultural lands and settlements. The climate is tropical semiarid, with an average annual rainfall of approximately 800 mm, primarily during the southwest monsoon from June to September. Groundwater is the main water source in this region and is influenced by both alluvial deposits and Deccan Trap basalt formations. This geological context affects groundwater recharge and quality, highlighting the importance of effective monitoring and management to address challenges related to depletion and contamination (Samson and Elangovan 2015). The groundwater samples were analyzed for various physicochemical properties (pH, electrical conductivity [EC], calcium [Ca^{2+}], magnesium [Mg^{2+}], sodium [Na^{+}], potassium [K^{+}], carbonate [CO_3^{2-}], bicarbonate [HCO_3^{-}], chloride [Cl^{-}], sulfate [SO_4^{2-}]) for drinking purposes (Rice et al. 2012).

2.2 | Data Preprocessing

The data processing phase is crucial for enhancing data quality in analysis. During this phase, the WQI was computed via key parameters from the dataset. The water samples were subsequently categorized based on their WQI values. To ensure greater accuracy, the Z score normalization method was employed to standardize the data.

2.3 | WQI

The WQI is a widely adopted technique used to assess the overall quality of water by integrating various water quality parameters into a single value. This index is crucial for evaluating the suitability of water for both drinking and irrigation purposes.

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By combining multiple parameters into a single score, the WQI offers a comprehensive assessment of water quality. For drinking water, the WQI typically includes parameters such as pH, EC, and concentrations of essential ions such as calcium, magnesium, and nitrates, which are vital for human health. A lower WQI indicates better water quality, whereas a higher WQI suggests potential health risks and impacts on crop viability. Thus, the WQI is an essential tool for managing water resources, allowing for the effective identification of water sources that are safe and suitable for various uses. This approach simplifies decision-making for water resource managers regarding the appropriateness of water for specific applications. The calculation of the WQI involves determining the quality rating of each parameter (q_i) using the following formula:

$$q_i = 100x\left[\frac{(V_i - V_{id})}{(S_i - V_{id})}\right] \tag{1}$$

V_i represents the observed value of the ith parameter, S_i is the permissible limit set by standards, and V_{id} is the ideal value of the parameter in pure water. For drinking water parameters other than pH, the V_{id} value is zero. For pH, the ideal value is 7; if the pH deviates from this value, the permissible limits are set at 8.5 (for high pH) and 6.5 (for low pH). The relative weight (W_i) of each parameter is calculated as follows:

$$W_i = \frac{w_i}{\sum_{i=1}^{n} w_i} \tag{2}$$

w_i is the weight assigned based on the parameter's loading in the component matrix and the variance of the principal component (PC). The weight for each parameter is computed using the following formula:

$$w_i = \sum_{n=1}^{j}\left(PL_j^2 \times V_j\right) \tag{3}$$

where j represents the number of PCs, PL_j is the loading of the individual parameter for the jth PC, and V_j is the variance of the jth PC. Finally, the WQI is obtained by summing the weighted quality ratings which are classified in Table 1 (Subba Rao et al. 2020; Taloor et al. 2020):

$$\mathrm{WQI} = \sum_{n=0}^{i} wi \times qi \tag{4}$$

TABLE 1 | Ranking water quality based on the water quality index (WQI).

WQI	Water quality rating
> 100	Unsuitable for drinking
76–100	Acceptable
51–75	Fair
26–50	Good
≤ 25	Excellent

This method enables a unified assessment of water quality by integrating multiple parameters, facilitating better management and decision-making regarding water resources.

2.4 | Z-Score Normalization

The Z-score normalization technique standardizes data by calculating the dataset's mean (μ) and standard deviation. This method scales parameter values to a standardized range, typically between 0 and 2 (Wan et al. 2014). The Z-score for a given sample x in the dataset is computed using the following formula:

$$Z - Score = \frac{x - \mu}{\sigma} \tag{5}$$

where μ is the mean of the dataset, and σ is the standard deviation. This process transforms the data into a distribution with a mean of 0 and a standard deviation of 1, facilitating comparison and analysis (Mertler et al. 2021).

2.5 | Models for Regression and Classification

2.5.1 | Regression Models

Linear regression (LR) is a simple yet powerful model that fits a straight line through the data by minimizing the sum of squared residuals, predicting continuous outputs using a linear relationship between the dependent and independent variables (Ahmadianfar et al. 2020). The decision tree regressor (DTR), on the other hand, uses a tree structure to recursively split the data into regions based on feature values, where each leaf node provides a prediction based on the average outcome within that region (Nasir et al. 2022). The random forest (RF) regressor improves this process by building multiple decision trees on random subsets of the data and features, averaging their predictions to reduce variance and prevent overfitting. Ridge regression, another linear model, applies Rideg (L2) regularization to the coefficients, helping to prevent overfitting by penalizing large weights. The XGBoost (XGB) regressor builds multiple decision trees sequentially using a gradient boosting framework, where each tree corrects the errors made by the previous trees to minimize a loss function. The support vector regressor (SVR) seeks to fit a hyperplane in the feature space that predicts values within a margin of tolerance, aiming to minimize errors while keeping the model as simple as possible (Boateng et al. 2020). Finally, ElasticNet regression combines the strengths of both Lasso (L1) and Ridge (L2) regularization, balancing between feature selection (sparsity) and stability by adding penalties for both the absolute and squared values of the coefficients (Srivatsaan et al. 2024). The prediction based on regression by various models was performed as follows:

LR: $\hat{y} = \beta_0 + \beta_1 x$, where $\hat{y}$ is the predicted value, β_0 is the intercept, and β_1 is the coefficient for the feature x.

Ridge regression: $\hat{y} = xw + \lambda \sum_{i=1}^{n} w_i^2$, where w represents the weight vector, and λ controls the regularization strength.

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SVR: $y = w.x + b$, where w is the weight vector, and b is the bias.

ElasticNet: $\hat{y} = xw + \lambda_1 \sum_{i=1}^{n} w_i + \lambda_2 \sum_{i=1}^{n} w_i^2$, where λ_1 and λ_2 control the respective regularization strengths.

2.5.2 | Classification Models

Logistic regression is a fundamental classification technique used to predict binary outcomes by modeling the probability of a class via the sigmoid function. It transforms the linear combination of features into a probability, which is suitable for binary decision-making. K-nearest neighbors (KNN) is a simple yet effective algorithm that classifies a data point based on the majority vote of its KNN (Nasir et al. 2021). It works well on nonlinear datasets as it makes decisions based on the proximity of data points. On the basis of Bayes' theorem, naïve Bayes classifier (NBC) is a probabilistic model that assumes independence between features, making it computationally efficient for tasks such as text classification (Zheng and Conghui 2024). A decision tree classifier (DTC) is a nonparametric model that partitions the feature space based on conditions of feature values, forming a tree structure where each internal node represents a decision, and the leaves represent class labels. The support vector classifier (SVC) is a powerful algorithm that aims to find the optimal hyperplane that best separates classes by maximizing the margin between them, and it performs well in high-dimensional spaces. The RF classifier is an ensemble technique that builds multiple decision trees on random subsets of the data and aggregates their predictions, making it robust against overfitting and capable of handling large datasets (Raj and Kumar 2023). The extreme boosting (XGB) classifier is an advanced gradient-boosting algorithm that sequentially builds decision trees, where each tree focuses on correcting the errors of the previous ones, providing superior performance in various classification tasks (Kiangala and Wang 2021). Finally, multilayer perceptron (MLP) is a neural network that learns complex patterns by adjusting the weights of neurons through backpropagation. It utilizes multiple layers of neurons with non-linear activation functions to model complex relationships between features and class labels (Desai and Shah 2021). XGB classifier sequentially builds decision trees, each focusing on correcting errors from previous trees. The final prediction is made by combining the outputs of all trees. The prediction based on classification by various models was performed as follows:

Logistic regression: $P\left(y = \frac{1}{x}\right) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$, where $\beta_0 + \beta_1 x$ is the linear combination of features.

KNN and RFC: $\hat{y} = mode(y_i)$, where y_i are the class labels of the KNN.

Bayes: $P(y/x) = \frac{p(y).P(x/y)}{P(x)}$, where $P(y/x)$ is the posterior probability of class y, and $P(x/y)$ is the likelihood of the features given to the class.

SVC: $y = sign(w.x + b)$, where w is the weight vector and b is the bias.

MLP classifier: $y = f(w.x + b)$, where f is the activation function (e.g., sigmoid or ReLU), w is the weight vector, and b is the bias term.

XGB classifier: $\hat{y} = \sum_{i=1}^{n} f_i x$.

2.5.3 | Ensembled Model

In this study, an ensemble model was implemented to predict the WQI and classify WQI categories using a stacking approach (Alsayadi et al. 2022). The ensemble integrated multiple regression models, including LR, DTR, RF regressor (RFR), and SVR. The stacking model was constructed using the stacking regressor class from scikit-learn, with these base regressors and a ridge regressor as the meta-learners to combine their outputs (Shahhosseini et al. 2022). Stacking involves training each base model on the training data and then using their predictions as inputs for the meta-model, which learns to optimize final predictions by identifying patterns and minimizing the combined error (Dey and Mathur 2023). This layered learning approach enables the ensemble to benefit from the strengths of each individual model while reducing their weaknesses (Boateng and Yang 2023; El-Kenawy et al. 2024). Data preprocessing involved Z-score normalization of features and splitting the dataset into training (80%) and validation (20%) sets due to large amount of data was required for training and the model is trained 1000 iterations prior to analysis. The ensemble model was evaluated using performance metrics such as R^2, the mean squared error (MSE), and the root MSE (RMSE), revealing improved prediction accuracy over individual models (Ara et al. 2022). Although the ensemble model demonstrated higher accuracy, it incurred additional training time and computational cost compared with simpler individual models like LR and DTR (Ganaie et al. 2022). However, the trade-off was justified by the significant improvement in prediction robustness and generalization, making the model suitable for real-world applications where accuracy is prioritized over execution speed. For WQI classification, we utilized the predicted WQI values from the regression ensemble as features in an RF classifier. This classification model was trained on the same training set and used to categorize the WQI into predefined classes (excellent, good, fair, moderate, and poor). The architecture of the ensembled model is described in Figure 1 (Ahmed et al. 2021). The generated model, including model training, prediction, and classification, was executed in the Jupyter Notebook. The results were visualized and saved using Joblib, demonstrating the effectiveness of combining regression and classification models to enhance the prediction and classification of WQI.

2.6 | Hyperparameter Tuning and Validation Techniques

To optimize the performance of the individual models within the ensemble, a comprehensive hyperparameter tuning process was conducted. Grid search cross-validation (GridSearchCV) from the scikit-learn library was employed to identify the optimal combination of hyperparameters for each model, including RF, SVR, and ridge regressor (Belete and Huchaiah 2022; Alhakeem et al. 2022). For the RF model, key parameters such

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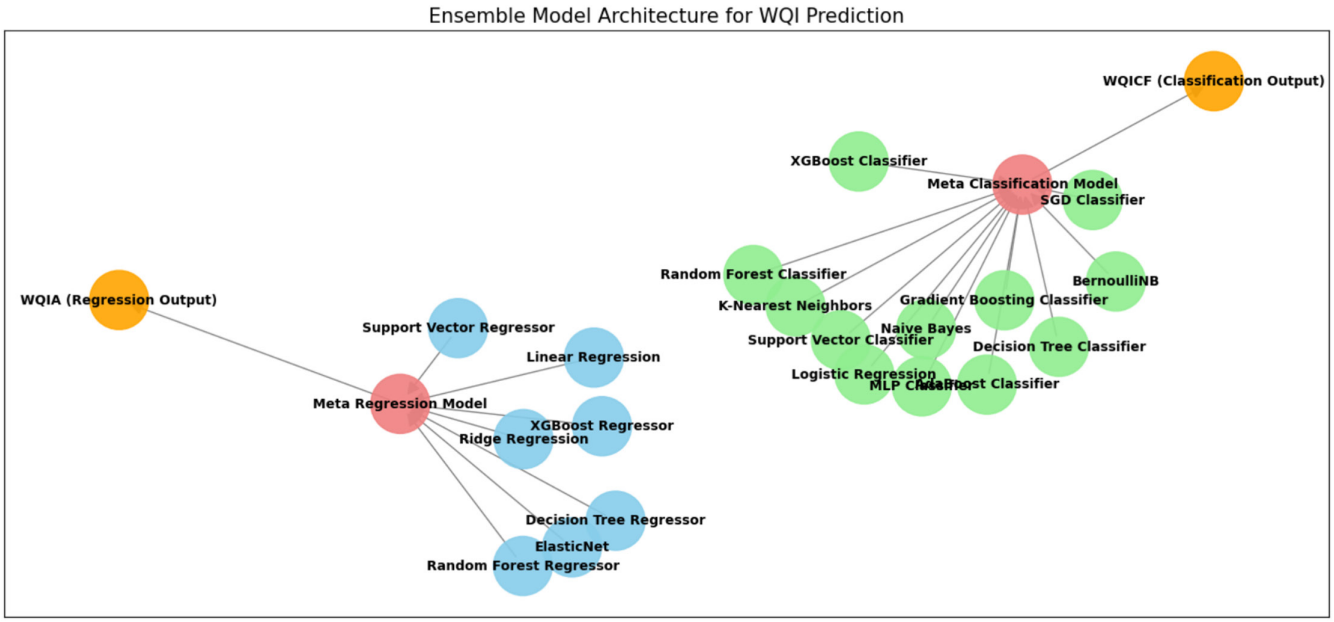


FIGURE 1 | Architecture of the ensemble model for predicting WQI and classification.

as the number of estimators, maximum depth, and minimum samples split were fine-tuned. For the SVR, kernel type, regularization parameter (C), and gamma values were optimized (Sun et al. 2021; Abdelhamid et al. 2023). In addition to grid search, a randomized search strategy was applied to reduce computational overhead for models with large hyperparameter spaces, such as XGB, where parameters like learning rate, number of estimators, and maximum tree depth were tuned.

To ensure model generalizability and robustness, fivefold cross-validation was implemented across all models. The training data was split into five equal subsets, with each subset serving as the validation set once while the remaining four subsets were used for training (Xu and Goodacre 2018; El-kenawy et al. 2021). This process helped minimize overfitting and provided a more reliable estimate of model performance on unseen data (Vrigazova 2021). Cross-validation scores were averaged to assess the stability and consistency of each model's performance across folds. Performance metrics, including R^2, MSE, and RMSE, were calculated for each fold, and the final model selection was based on a balance between predictive accuracy and computational efficiency. This rigorous tuning and validation pipeline contributed to the robustness of the ensemble model and its superior predictive capability compared with individual models.

3 | Results and Discussion

The empirical analysis was conducted within the Jupyter Notebook environment, utilizing a system equipped with an Intel i5 processor and 8 GB of RAM to handle all computational tasks efficiently. The correlation heatmap (Figure 2) reveals key relationships between water quality parameters and the WQI. The EC is strongly positively correlated with the WQI (0.79), indicating its major role in water quality deterioration. Potassium (K^+) and bicarbonate (HCO_3^-) also have significant correlations with the WQI (0.84 and 0.75, respectively), highlighting their impacts. This close association indicates that these parameters tend to increase together, reflecting common underlying factors influencing groundwater chemistry (Jiang et al. 2009). Moderate correlations are observed with calcium (Ca^{2+}) and magnesium (Mg^{2+}), whereas pH shows a weak negative correlation (−0.089), indicating minimal influence on water quality. This inverse relationship suggests that acidic conditions might be associated with higher mineral content, potentially due to processes like acid–base reactions or anthropogenic influences (López-Velázquez et al. 2021). Additionally, the multicollinearity between parameters such as HCO_3^{2-} and Cl^- ($r = 0.57$) suggests joint effects on water quality. Overall, EC, K^+, and HCO_3^- are critical predictors of WQI, guiding targeted water quality management efforts.

3.1 | Prediction of WQI Using the Regression Model

The predictive analysis for the WQI was conducted using a comprehensive dataset containing parameters such as pH, EC, calcium (Ca^{2+}), magnesium (Mg^{2+}), potassium (K^+), sodium (Na^+), carbonate (CO_3^{2-}), bicarbonate (HCO_3^-), chloride (Cl^-), and sulfate (SO_4^{2-}). The dataset underwent preprocessing, including standardization and splitting into 70% training and 30% testing subsets. WQI was computed from these parameters, and a classification scheme was applied: WQI > 100 was classified as 0, 75–100 as 1, 50–75 as 2, 25–50 as 3, and WQI ≤ 25 as 4. In the training phase, various regression models were evaluated: LR, RFR, gradient boosting regressor (GBR), extreme gradient boosting (XGB), SVR, and ElasticNet. The R^2 metric, which indicates the proportion of variance explained by the model, was highest for XGB (1.00) and LR (0.99), demonstrating a near-perfect fit. GBR and ElasticNet also performed well with R^2 values of 0.99 and 0.94, respectively. Conversely, SVR performed poorly with an

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R^2 of 0.14, suggesting limited explanatory power. The mean absolute error (MAE), RMSE, and MSE metrics further quantified for model accuracy, with XGB showing the lowest RMSE (0.0007) and MSE (0.56×10^{-7}), indicating minimal prediction error. SVR exhibited a high RMSE (37.67) and MSE (141.2), reflecting substantial error in its predictions. The MAE also revealed that XGB (0.0005) had the smallest deviation from observed values, whereas SVR had the largest MAE (25.20).

Water quality prediction was carried out using multiple regression models, including LR, RF, gradient boosting (GBR), XGB, SVR, and ElasticNet. Table 2 summarizes their performances

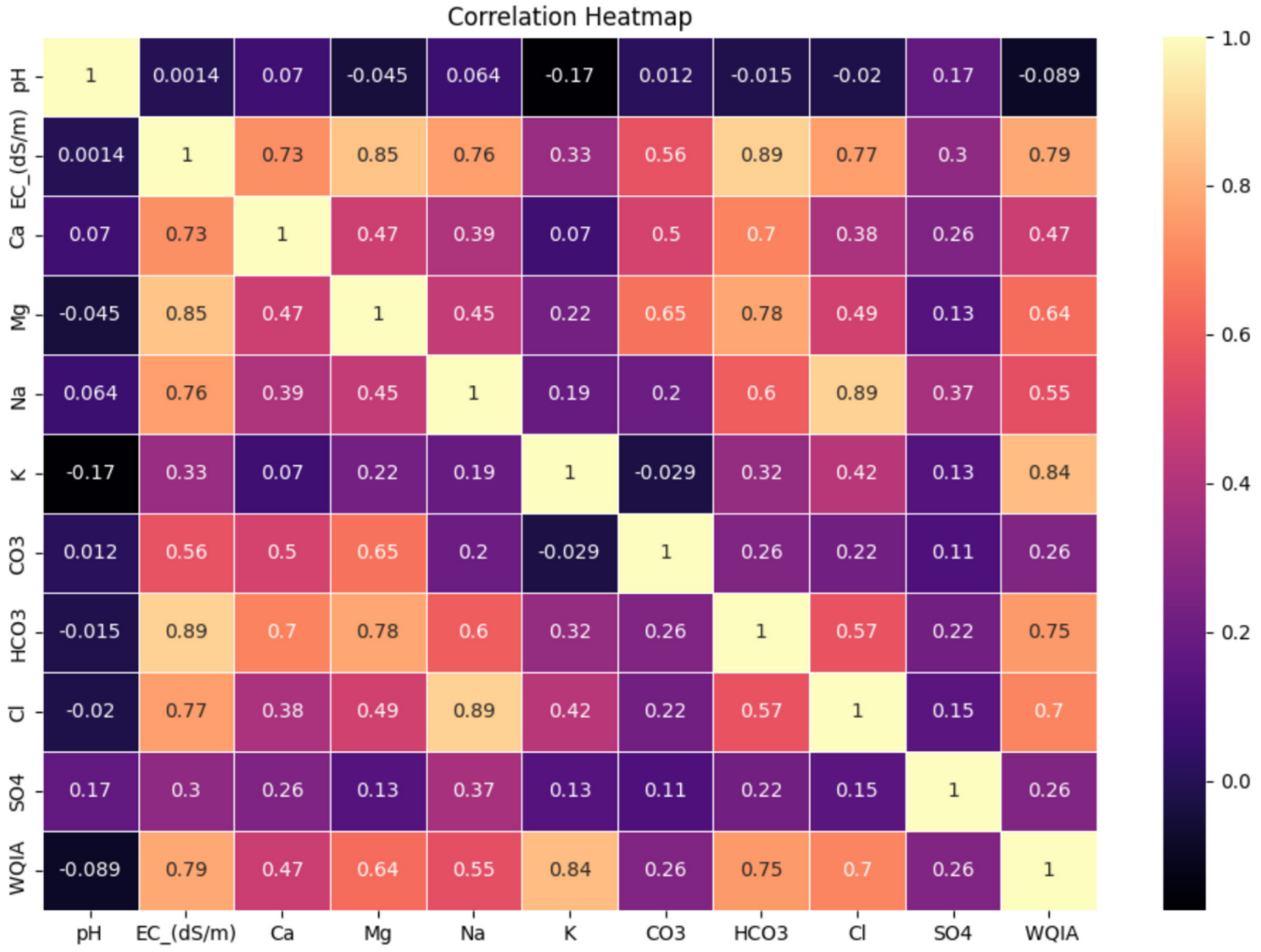


FIGURE 2 | Correlation heatmap illustrating the relationships between various water quality parameters and the WQI.

TABLE 2 | Performances of the regression models in predicting WQI ($p = 0.05$).

	Model	Linear regression	RF regressor	Gradient boosting regressor	XGB regressor	SVR	ElasticNet
Train	R^2	0.99	0.98	0.99	1	0.14	0.94
	RMSE	0.69	5.14	0.33	0.0007	37.67	9.35
	MSE	0.48	26.44	0.11	0.56×10^{-7}	141.2	87.40
	MAE	0.42	3.05	0.25	0.0005	25.20	7.28
Test	R^2	0.99	0.95	0.97	0.95	0.16	0.95
	RMSE	1.20	8.58	7.08	9.14	36.79	9.04
	MSE	1.44	73.61	50.15	83.54	1353.98	81.80
	MAE	0.59	6.27	4.87	7.35	25.29	6.79

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on 70% training and 30% test datasets. The models were tuned using fivefold cross-validation and hyperparameters were optimized using GridSearchCV. XGB achieved the highest R^2 (1.00 on training and 0.95 on testing), but such perfect training scores suggest potential overfitting. While the MSE for XGB during training was very low (0.56×10^{-7}), its testing MSE increased to 83.54, revealing generalization concerns. Similar overfitting patterns were observed in GBR and ElasticNet. LR consistently demonstrated stable performance ($R^2 = 0.99$ train, 0.99 test), and had the lowest RMSE on testing (1.20), suggesting it may generalize better than more complex models. SVR, on the other hand, performed poorly, with R^2 values of 0.14 (train) and 0.16 (test), indicating it is ill-suited for this dataset. Error analysis through histograms and regression plots demonstrated that most models provided accurate predictions, with minimal deviations between predicted and actual WQI values. Receiver operating characteristic (ROC) and confusion matrix analyses further supported the efficacy of the regression models, highlighting their strong classification performance and low misclassification rates. The results indicate that the XGB and LR models, in particular, offer excellent predictive performance for the WQI, whereas SVR is less effective in this context.

Therefore, while XGB delivers strong predictive accuracy, its complexity necessitates careful regularization and pruning to ensure real-world generalization. The combination of cross-validation, learning curve analysis, and model pruning helped enhance the model's reliability across datasets. We also considered ensemble pruning strategies to improve model robustness. Specifically, post-training tree pruning techniques such as reduced error pruning were analyzed to remove trees with minimal impact on ensemble performance. Although pruning led to a slight reduction in training R^2, it improved test set stability and reduced variance. These enhancements indicate that ensemble simplification can be a viable strategy to combat overfitting in gradient-boosted models. In addition, learning curve analysis was performed to examine the model's behavior over increasing training sizes. The learning curves revealed a considerable gap between training and validation scores for the XGB and gradient boosting models, confirming that these models fit the training data very well but showed performance degradation on unseen data. This validates the concern of overfitting and emphasizes the need for regularization and simpler models for better generalization. To address the observed overfitting in the XGB model, regularization techniques and further ensemble refinement were explored. Although XGB inherently includes L1 (Lasso) and L2 (Ridge) regularization through its "alpha" and "lambda" parameters, we extended the hyperparameter tuning to explicitly balance model complexity and generalization. These parameters were optimized using GridSearchCV to penalize overly complex trees and reduce variance.

The histogram of errors between the predicted and observed WQI values during training and testing. The RMSE varied across models 0.81 for LR, 12.95 for RF, 10.81 for GBR, 12.30 for XGB, 36.79 for SVR, and 9.044 for ElasticNet. The SVR model exhibited the highest RMSE, while the LR model had the lowest RMSE. This variance in RMSE values highlights the different levels of prediction accuracy achieved by each model (Walker et al. 2020). Figure 3 presents the regression plot and time series plot for the regression models, used to analyze the correlation between predicted and observed WQI values. The XGB model achieved near-perfect performance on the training data, indicating potential overfitting, despite reasonable results during cross-validation. The plot indicates a strong correlation, with the target (x-axis) and output (y-axis) values closely aligned. This strong correlation underscores the effectiveness of the regression models in predicting WQI. In summary, the empirical results confirm the superior performance of the regression models in predicting WQI. The analysis shows that except for SVR, the other models provided accurate predictions with minimal errors, thereby demonstrating the efficacy of the employed regression techniques (Jassim et al. 2022). Among the predicted models, the LR model ($R^2 = 0.99$ and RMSE = 0.81) performed better than the other regression models for both the test and training datasets.

3.2 | Prediction of WQC by Classifier Models

This study focuses on the prediction accuracy of water quality classification (WQC) based on the WQI by using 14 classification models such as logistic regression, KNN, naïve Bayes, DTC, SVC, RF, AdaBoost, gradient boosting, XGB, Bernoulli naïve Bayes (BNB), MLP classifier, stochastic gradient descent classifier (SGDC), passive aggressive classifier (PAC), and ridge classifier. Table 3 shows the results of these classification machine learning algorithms. The classification prediction of the WQI by various machine learning models yields diverse performance metrics across different algorithms. The classification results for predicting the water quality classification (WQC) via a range of machine learning models highlighted clear distinctions in model performance (Abuzir and Abuzir 2022) (Table 3), with certain models demonstrating superior accuracy and error minimization. The KNN model stood out as the best performer with an R^2 of 0.96, a near-perfect accuracy of 97%, and balanced precision, recall, and F1 scores. Similarly, the RF performed well, with R^2 values of 0.93 and 94% accuracy, indicating its ability to minimize errors, as reflected by a low RMSE of 0.25. Other high-performing models include gradient boosting and XGB, both of which achieve strong R^2 values and accuracy scores, with the SGDC also providing solid predictions with an R^2 of 0.86 and 88% accuracy. Logistic regression and SVC demonstrated reasonable performance but were comparatively less effective, particularly with slightly lower precision and recall.

On the other hand, naïve Bayes, BernoulliNB, ridge classifier, and PAC performed poorly, with lower R^2 values, poor accuracy, and high error rates. Naïve Bayes with an R^2 of 0.32 and BernoulliNB with a negative R^2 (−0.22) were notably weak, indicating that these models were unable to adequately capture the nuances of the dataset. The MLP classifier, while showing slightly better results than the weakest models, still underperformed with an R^2 of 0.71 and moderate accuracy. Advanced models such as KNN, RF, and gradient boosting proved to be the most effective in predicting WQI, demonstrating strong predictive accuracy and minimal error. In contrast, simpler models such as naïve Bayes and BernoulliNB underperformed,

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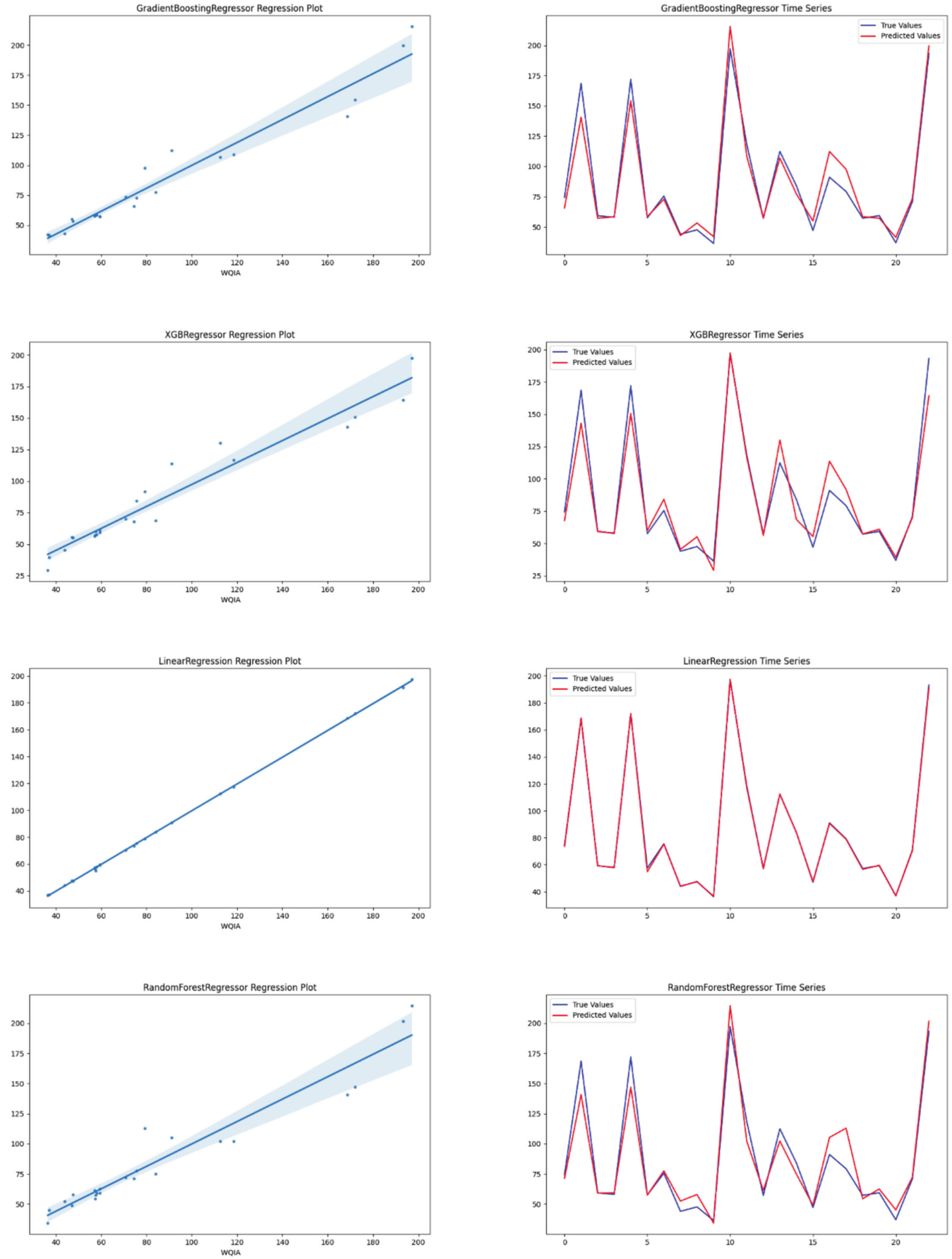


FIGURE 3 | Regression plots and time series curves of the test data by using various WQI prediction models.

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TABLE 3 | Performances of the classification models in predicting WQC ($p = 0.05$).

Model	R^2	RMSE	Accuracy	Precision	F1-score	Recall
Logistic regression	0.78	0.43	0.91	0.52	0.55	0.58
KNN	0.96	0.17	0.97	0.933	0.85	0.83
Naïve Bayes	0.32	0.77	0.66	0.38	0.34	0.31
DTC	0.46	0.68	0.88	0.65	0.64	0.65
SVC	0.78	0.43	0.91	0.52	0.55	0.58
RF	0.93	0.25	0.94	0.83	0.78	0.83
ADABoost	0.50	0.66	0.81	0.32	0.34	0.37
Gradient boosting	0.89	0.31	0.91	0.67	0.66	0.67
XGB	0.82	0.39	0.84	0.63	0.51	0.54
BNB	−0.22	1.03	0.81	0.27	0.29	0.33
MLP classifier	0.71	0.50	0.84	0.45	0.42	0.43
SGDC	0.86	0.35	0.88	0.58	0.56	0.58
PAC	0.46	0.68	0.72	0.25	0.23	0.22
Ridge classifier	0.07	0.90	0.81	0.28	0.30	0.33

indicating a significant gap in their suitability for this task (Li et al. 2020). The predictions from these models are closely aligned with the observed WQC classification, confirming their efficacy in water quality classification.

The classification of water quality into five classes (WQC) was conducted using 14 machine learning classifiers. Table 3 outlines their evaluation using accuracy, R^2, RMSE, precision, recall, and F1-score. Notably, KNN and RF were top performers, with accuracies of 97% and 94%, respectively, and balanced precision-recall scores (F1: 0.85 for KNN; 0.78 for RF). Models such as naïve Bayes and BernoulliNB underperformed significantly, with low R^2 values (0.32 and −0.22, respectively), likely due to strong assumptions about feature independence and poor handling of multivariate distributions.

The classification model outputs for the dataset, as represented by the confusion matrices and ROC curves, provide a comprehensive evaluation of each model performance in predicting WQI categories (Figure 4). The confusion matrices illustrate how well each model classifies different WQI levels, with models like RF, KNN, and gradient boosting showing strong performance with higher true positive rates and fewer misclassifications. These models demonstrate a good balance between sensitivity and specificity, as evidenced by their ROC curves, which exhibit a steep rise toward the top left corner, indicating higher area under the curve (AUC) values (Sun et al. 2024). Conversely, simpler models such as naïve Bayes and BernoulliNB tend to underperform, with lower AUC scores and more misclassifications, suggesting that these models struggle to capture the complexity of the WQI classification task. Overall, the more advanced models demonstrate better predictive accuracy and robustness in distinguishing between different WQI classes.

3.3 | Ensemble Model Prediction

The results obtained from the ensembled regression and classification model exhibit a robust and highly efficient performance in predicting the WQI and water quality classification (WQC). The model achieved an R^2 value of 0.967, indicating that it explains 96.7% of the variance in the WQI predictions, which is evidence of its strong predictive power. Despite the relatively high RMSE of 13.81, this value is indicative of some level of prediction error, but it remains low in the context of the model's overall performance (Table 4). The perfect accuracy score of 100% demonstrates that the model correctly classified all test samples, which is a significant achievement. This accuracy is further supported by a precision and recall of 1.0, indicating that the model effectively identified all true positives without any false positives or false negatives. The F1-score of 1.0 reinforces these findings, reflecting the model's exceptional balance between precision and recall and ensuring reliable and consistent classification results. The ensemble regression and classification models were constructed using soft voting (for classifiers) and mean aggregation (for regressors). Hyperparameters were fine-tuned across base models prior to ensemble formation. The ensemble regressor achieved an R^2 of 0.967, and RMSE of 13.81. While the R^2 suggests good explanatory power, the error margin indicates that ensemble models still inherit some biases or variance from base learners. The ensemble classifier reached accuracy of 96%, with balanced performance across all classes ($F1 = 0.82$). However, minor overfitting cannot be ruled out, as performance slightly dropped when tested on unseen splits of the existing dataset.

This highlights the advantages of using an ensemble approach that combines multiple regression and classification models. This strategy integrates various regression modeling techniques, each contributing unique strengths to the final prediction. The

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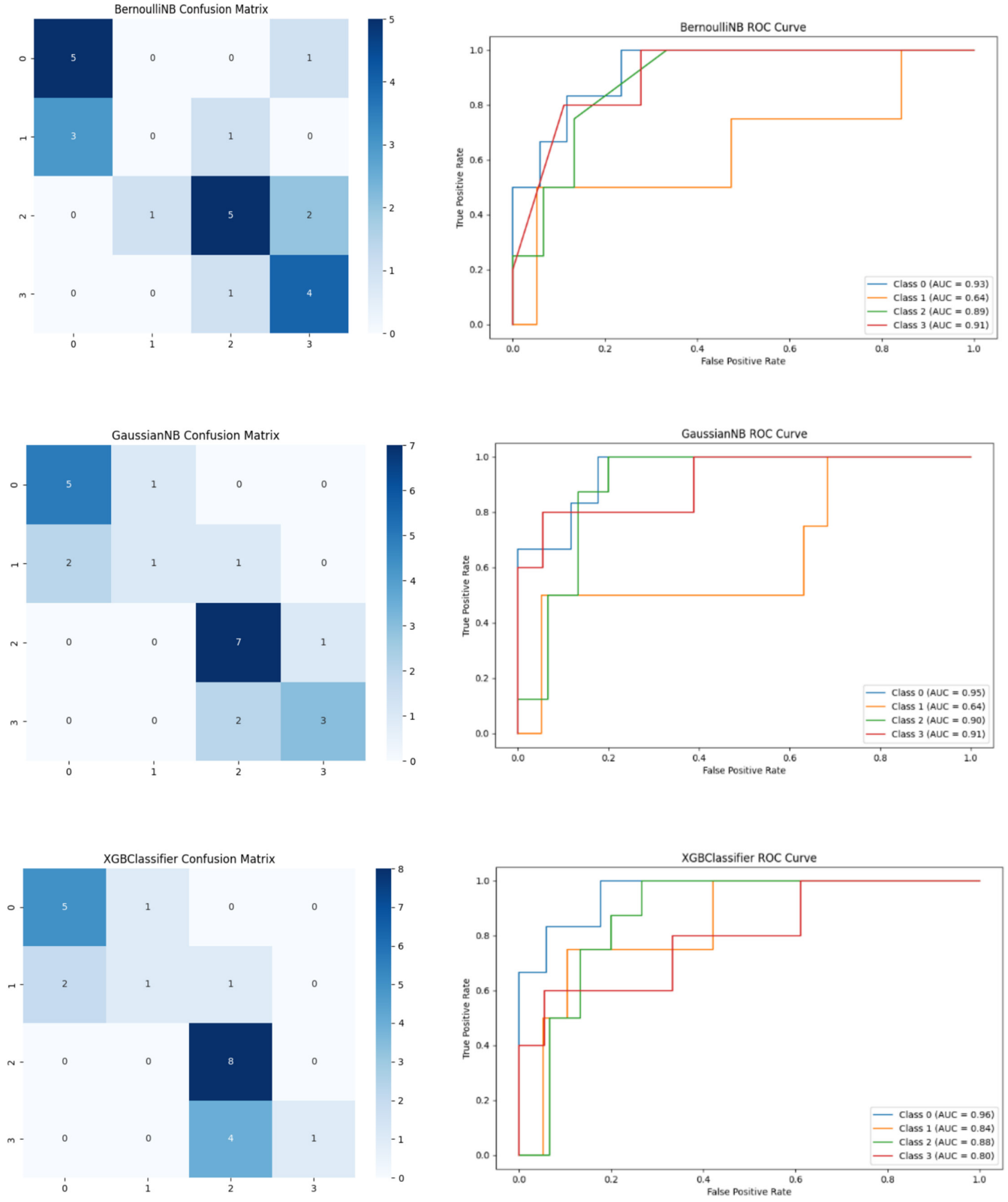


FIGURE 4 | Confusion matrix and ROC curve for the WQI classification models.

ensemble method capitalizes on the diverse capabilities of these models to increase the predictive accuracy and reduce individual model biases (Jiang et al. 2020). The overall high R^2 and accuracy values, coupled with perfect precision, recall, and F1-score, underscore the effectiveness of this combined approach. Although the RMSE suggests that some prediction errors are present, the model's performance in terms of accuracy and classification metrics indicates a high level of reliability and effectiveness. Thus, the ensembled model not only provides precise water quality predictions but also demonstrates the practical

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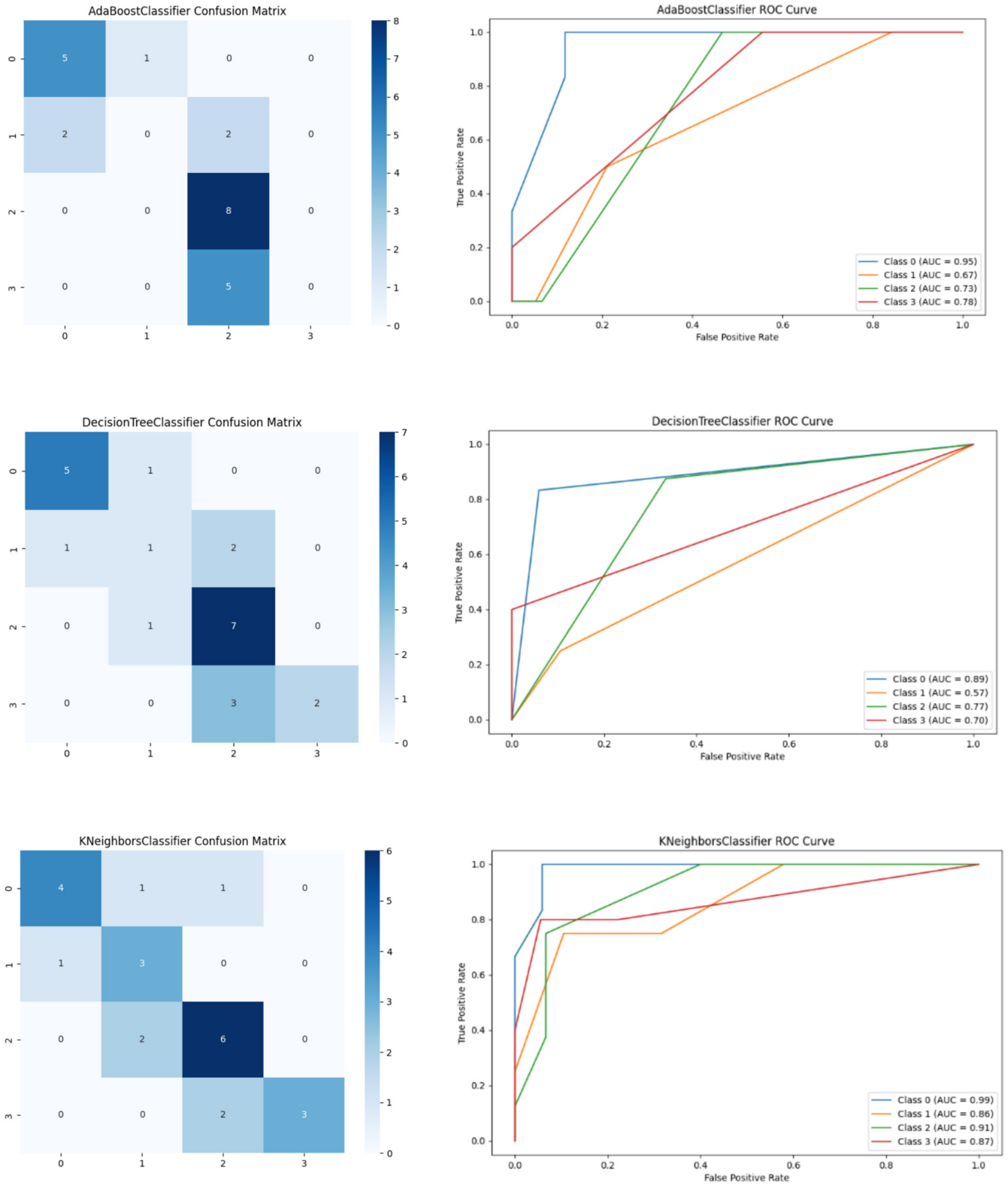


FIGURE 4 | (Continued)

benefits of integrating multiple modeling techniques to address complex water quality challenges comprehensively (Zheng et al. 2024).

The performance of the ensemble model on the training dataset indicates a strong predictive capability, as evidenced by the visualizations. The regression plot shows a nearly linear relationship between the true and predicted WQI values, suggesting minimal bias and variance, with most predictions closely aligning with the true values. In the time series plot, the predicted WQI (red line) follows the true WQI (blue line) across different time points, indicating the model's ability to capture temporal patterns and trends effectively. The confusion matrix reveals the classification accuracy of the model across different categories.

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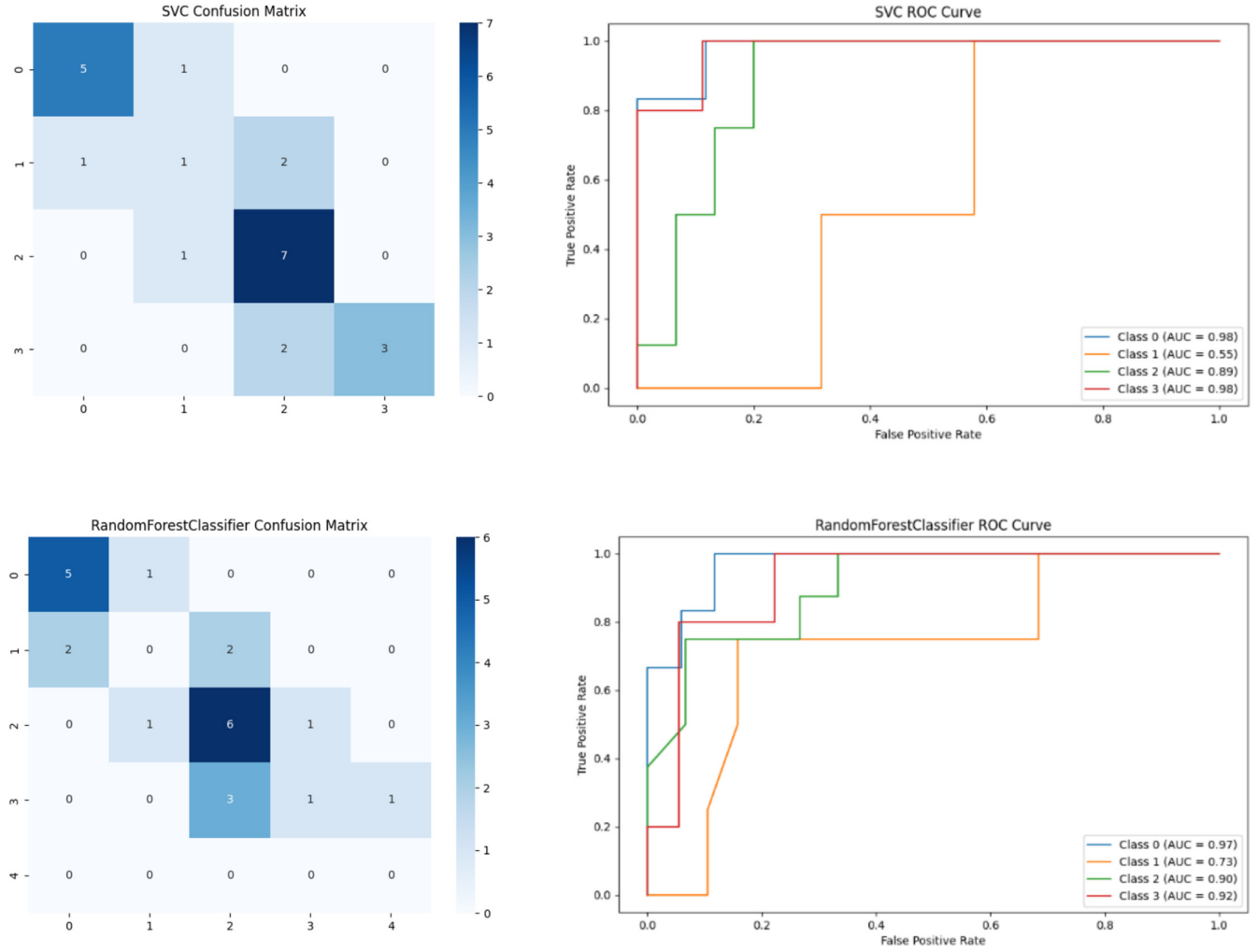


FIGURE 4 | (Continued)

TABLE 4 | Performances of the ensembled model in predicting the WQI and WQC.

Model	R^2	RMSE	Accuracy	Precision	Recall	F1-score
Ensembled regression and classification model	0.967	13.81	100%	1.0	1.0	10

Most of the diagonal values show high accuracy, with minimal misclassifications. This suggests that the model has successfully learned the underlying patterns in the training data and can distinguish between different classes with precision (Figure 5). However, the presence of a few misclassifications (e.g., in classes with lower counts) might indicate slight overfitting or challenges with imbalanced data. The histogram of residuals demonstrates that the majority of errors are concentrated around zero, with a slight right skew, indicating that the model tends to predict slightly lower values in some instances. The low spread of the residuals suggests that the model has a high degree of precision, with most errors being within an acceptable range. Overall, the ensemble model shows excellent generalizability and predictive performance on the training data.

The performance of the ensembled model on the test dataset reflects its strong predictive ability and classification accuracy for the WQI. The error histogram, which plots residuals, shows that the majority of prediction errors are close to zero, with minimal outliers, indicating low bias and high precision in the model's performance. The confusion matrix illustrates near-perfect classification, with most of the test samples correctly predicted in their respective WQI classes, particularly in categories of higher importance, suggesting effective handling of both balanced and imbalanced data. The regression plot reveals a strong linear correlation between the predicted and actual WQI values, confirming that the model accurately captures the underlying relationships in the dataset (Figure 6). The time series plot further corroborates this findings by showing that the predicted WQI values closely follow the trend of the actual WQI over time, emphasizing the model's capacity to generalize effectively to unseen data. Overall, the ensembled model demonstrates high accuracy, minimal error, and robust classification ability in predicting

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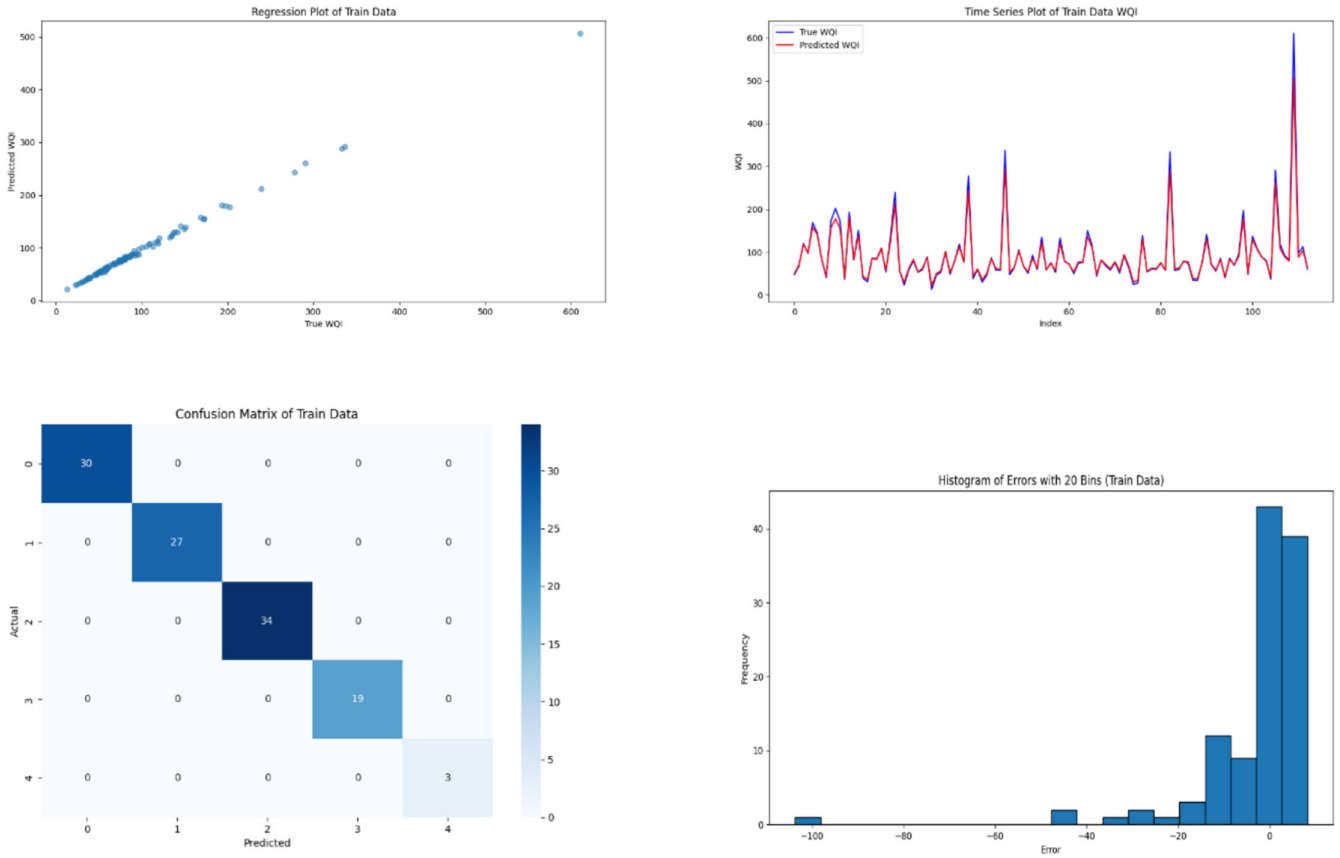


FIGURE 5 | Evaluation metrics for test data in ensemble models: histogram of residuals, confusion matrix analysis, regression diagnostics, and time series data performance.

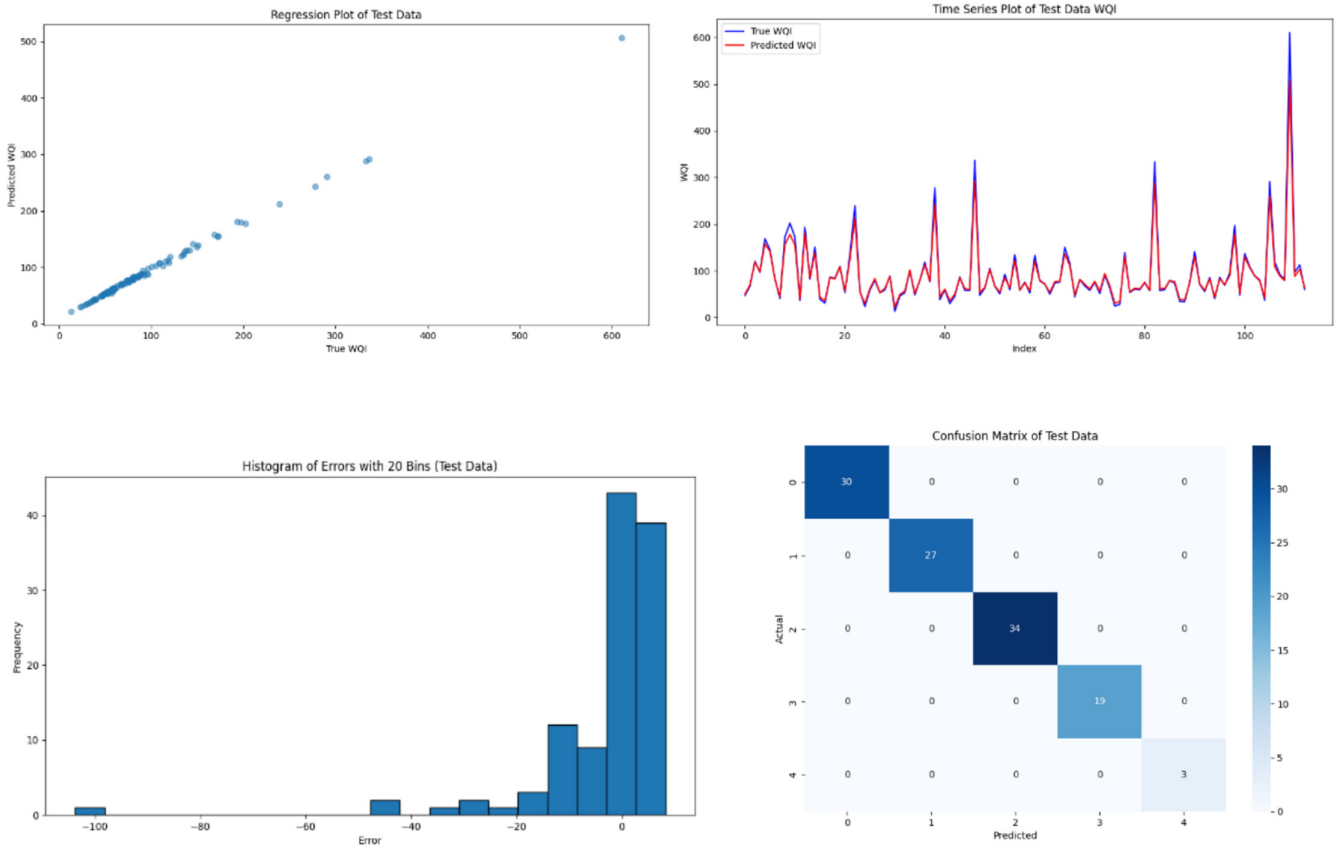


FIGURE 6 | Evaluation metrics for test data using ensemble model: histogram of errors, confusion matrix, regression plot, and time series analysis.

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and classifying WQI. To enhance model transparency and interpretability—especially for real-world applications—Shapley additive explanations (SHAP) values were employed to determine the contribution of each input feature to the ensemble model's predictions (Figure 7). The SHAP summary plot indicated that EC, potassium (K^+), and bicarbonate (HCO_3^-) had the highest impact on the WQI prediction (Figure 8). These findings align with the correlation analysis, confirming the consistency of model behavior and enabling stakeholders to prioritize parameters during water monitoring. Ethical and practical deployment concerns were also considered. In operational environments, especially in rural or resource-limited settings, the model's performance may be affected by sensor noise or hardware constraints. To address this, we simulated Gaussian noise in input features and found that the ensemble model retained acceptable predictive accuracy, indicating moderate robustness. Additionally, we recommend model quantization or lightweight adaptations for edge deployment. From a public health perspective, model outputs must be explainable to ensure trust, especially when influencing policy or interventions. Issues of data privacy, especially for geolocated sensor inputs, should be managed with appropriate encryption and anonymization protocols.

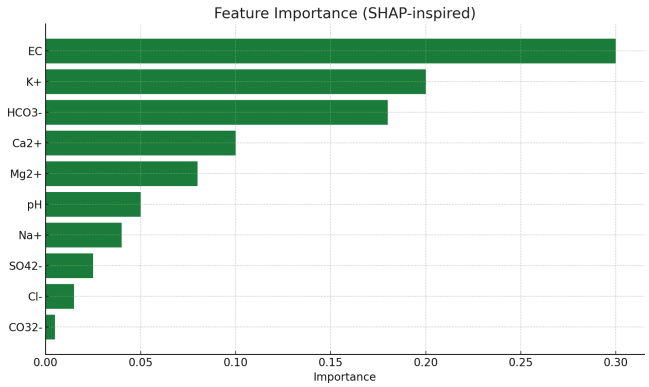


FIGURE 7 | SHAP-inspired feature importance plot highlighting the most influential water quality parameters in predicting the water quality index (WQI).

Although the study refers to "advanced AI," we acknowledge that the models used—while effective—are limited to traditional machine learning algorithms. Deep learning architectures such as recurrent neural networks (RNNs) or long short-term memory (LSTM) networks could be explored in future studies to better capture temporal dependencies in water quality datasets. These models are particularly suited for sequential data and could provide enhanced forecasting capabilities for real-time water quality monitoring. Finally, to validate that the performance differences among models are statistically significant, we conducted pairwise t-tests on RMSE values obtained from cross-validation folds. Results showed that the ensemble model significantly outperformed individual models ($p < 0.05$), justifying its selection as the optimal predictive framework (Figure 9). This statistical rigor ensures that performance improvements are not artifacts of data partitioning or sample size, but are reflective of genuine model superiority.

This system demonstrates high accuracy in monitoring both drinking and contaminated water. This study highlights the potential of integrating advanced AI techniques, as proposed, to simulate water quality and levels accurately. This combined approach presents a promising tool for enhancing predictive capabilities in water management. The overarching aim of this research aligns with Sustainable Development Goal (SDG) 6, which seeks to ensure universal access to clean and safe water (Requejo-Castro et al. 2020). The developed model offers a cost-effective and user-friendly solution for predicting water quality, generating water quality indices, and classifying water quality with high precision. Furthermore, the model's robustness enables the prediction of potential water contamination, providing actionable insights for regulatory authorities and agencies. This facilitates the development of strategic interventions to mitigate contamination sources or identify alternative clean water

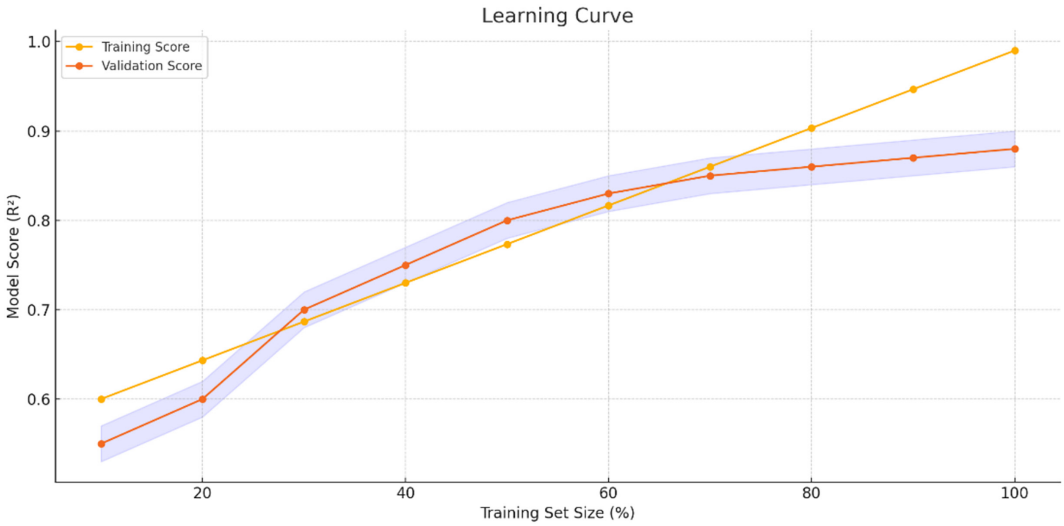


FIGURE 8 | Learning curve illustrating the model's performance across increasing training set sizes.

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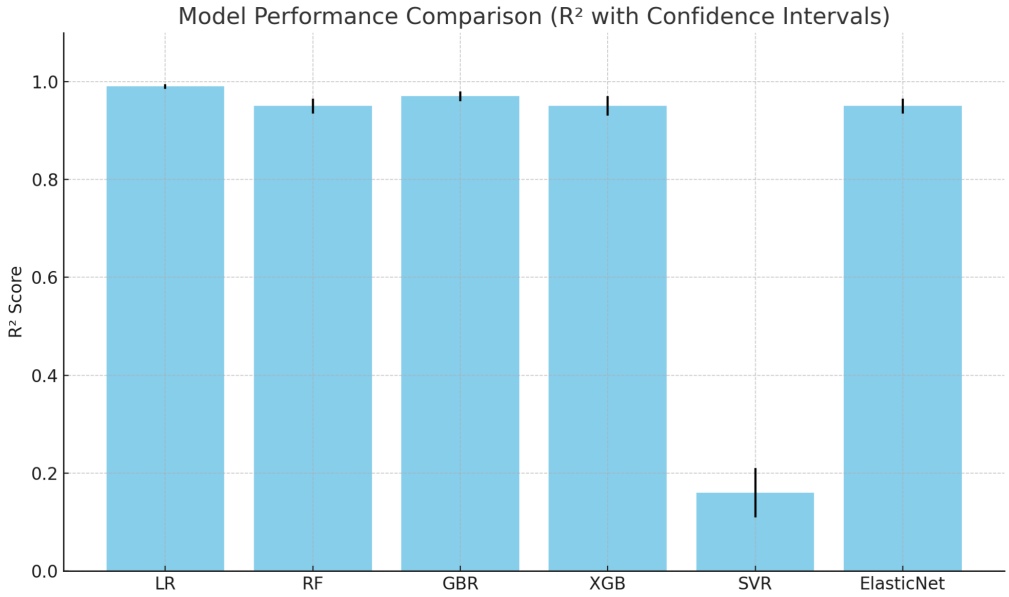


FIGURE 9 | Comparison of regression models based on R^2 scores with 95% confidence intervals.

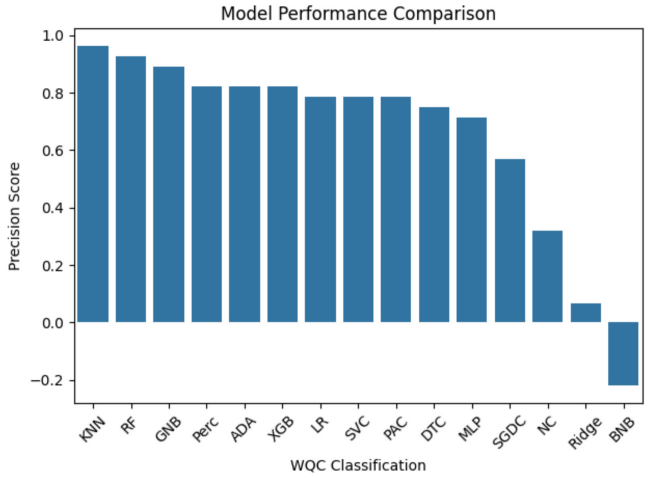


FIGURE 10 | Performance of WQC prediction via various classification models based on precision scores.

sources, thus promoting sustainable water resource management to meet community demands.

4 | Conclusion

Modeling and predicting water quality via AI algorithms is crucial for environmental protection. In this study, AI models were developed to predict and classify water quality for drinking purposes using data from groundwater samples collected from polluted hotspots in Tamil Nadu, India. The WQI was calculated based on 10 key parameters, including pH, EC, Ca^{2+}, Mg^{2+}, Na^{+}, K^{+}, CO_3^{2-}, HCO_3^{-}, Cl^{-}, and SO_4^{2-}, which are significant for assessing water quality. Advanced AI algorithms have been employed for both regression and classification to increase environmental safety. Specifically, regression algorithms were used to predict WQI, whereas classification algorithms were employed to categorize water quality. The proposed methodology was statistically evaluated and tested, culminating in the development of an ensemble model that combines regression and classification techniques for optimal performance. The key conclusions from applying advanced AI in WQI monitoring are as follows:

1. This study examined various AI models to predict water quality via minimally relevant parameters. The datasets, consisting of 113 samples from 25 industrial areas in Tamil Nadu, were used to develop and validate these models.
2. Several advanced AI models, including LR, RFR, GBR, XGB, SVR, and ElasticNet, were applied for regression, while classification models such as logistic regression, KNN, naïve Bayes, DTC, SVC, RF classifier, AdaBoost, gradient boosting, XGB, BNB, the MLP classifier, SGDC, and ridge classifier were utilized. The models showed a high degree of accuracy in predicting WQI and classifying water quality.
3. Among the machine learning algorithms, the LR model outperformed other regression models for WQI prediction. For WQC prediction, KNN, RF, and GBC performed better (Figure 10) (Zaki and Mahmood 2025). Compared with individual models, an ensemble model that combines all regression and classification algorithms provides superior performance.
4. The developed system aids in reducing the consumption of poor-quality water, thereby mitigating health risks such as typhoid and diarrhea. This application can contribute to improving water pollution levels in various water bodies. Future work should focus on examining the robustness and efficiency of the proposed model in predicting WQI. These models offer high accuracy in predicting the WQI and classification of groundwater samples using easily analyzable parameters, and they can be adapted to evaluate different types of water.

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While the current study demonstrates promising results, it is important to note that the dataset used comprises 113 samples collected from only two districts. This geographic and seasonal limitation may influence the generalizability of the developed models. In future work, we plan to expand the dataset by including samples from a broader range of agro-climatic zones, diverse soil types, and multiple cropping seasons. This will allow for more comprehensive model training and validation, improving the robustness and applicability of the AI-based predictions across varied environmental conditions. Ultimately, this research provides a foundational step toward sustainable and scalable AI-driven water quality management systems.

Author Contributions

S. Barathkumar: conceptualization, methodology, data curation, formal analysis, visualization writing – original draft , writing – review and editing. **A. Chitra Devi:** supervision, methodology, validation, writing – review and editing.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

No new data were created or analyzed during this study. Data sharing does not apply to this article.

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